

PROJECT REPORT ON

**PollutionNav: Spatial AQI Interpolation and Pollution Optimal
Routing in Mumbai**

SUBMITTED TO

SVKM'S NMIMS (DEEMED-TO-BE UNIVERSITY) IN

PARTIAL FULFILLMENT OF THE DEGREE OF

**BACHELOR OF SCIENCE
IN
DATA SCIENCE**

BY

PRATYAKSH BHUTANI A007

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Under the Guidance of

SHRADDHA SARODE

April , 2026



**NILKAMAL SCHOOL OF MATHEMATICS,
APPLIED STATISTICS & ANALYTICS**

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April , 2026



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Mumbai - 400056

CERTIFICATE

This is to certify that work described in this thesis entitled “**PollutionNav: Spatial AQI Interpolation and Pollution Optimal Routing in Mumbai**” has been carried out by **Mr Shitiz Kumar Gupta, Mr. Pratyaksh Bhutani, Mr. Husain Bootwala, Mr. Anas Shaikh** and **Mr. Keegan Nunes** under my supervision. I certify that this is his/her Bonafide work. The work described is original and has not been submitted for any degree to this or any other University.

Date:

Place: Mumbai

Supervisor

(Prof. Shraddha Sarode)

Year: 2026

Statement by the Candidate

This is to submit that this written submission in my report entitled "**PollutionNav: Spatial AQI Interpolation and Pollution Optimal Routing in Mumbai**" presents my ideas in my own words, and where others' ideas or words have been included, I have adequately cited and referenced the sources.

I also declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my submission.

I understand that any violation of the above will be cause for disciplinary action by the school and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

Mr. Shitiz Gupta, Mr. Keegan Nunes, Mr. Husain Bootwala, Mr. Anas Shaikh, Mr. Pratyaksh Bhutani

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TABLE OF CONTENTS

ABSTRACT	1
1. INTRODUCTION	2
2. RATIONALE	3
3. LITERATURE REVIEW	4
4. OBJECTIVES	5
5. DATA PREPARATION	6
5.1 Data Source and Extraction	6
5.2 Preprocessing Pipeline	7
5.3 AQI Computation	9
5.4 Correlation Analysis	10
6. METHODOLOGY	12
6.1 System Architecture and Block Diagram	12
6.2 CPCB Air Quality Index Computation	12
6.3 Live Data Ingestion via OpenAQ v3 API	15
6.4 Coordinate Reference Systems	16
6.5 Spatial Interpolation of the AQI Surface	16
6.6 Interpolation Evaluation	19
6.7 Pollution-Aware Route Planning	20
6.8 Streamlit Web Application	24
6.9 Land Use Regression (LUR) Exploratory Modelling	26
7. RESULTS AND DISCUSSION	30
7.1 Interpolation Method Evaluation LOSO Cross-Validation Results	30
7.2 Application Output Results	33
7.3 Land Use Regression Results and Discussion	37
7.4 Synthesis and Overall Assessment	39
8. SUMMARY AND CONCLUSION	40
9. APPENDIX	41
REFERENCES	42

ABSTRACT

Urban air pollution poses a severe public health threat across Indian cities, with Mumbai's 12.5 million daily commuters chronically exposed to PM_{2.5}, PM₁₀, NO₂, CO, and SO₂ from vehicular traffic, construction, and industrial activity. No existing tool enables residents to make navigation decisions based on cumulative pollution exposure — Google Maps optimises exclusively for time, dedicated AQI applications provide station-level readings without routing integration, and physical AQI boards have been documented showing readings differing by over 80 AQI units at adjacent junctions. This project presents PollutionNav, a real-time pollution-aware route planning system for Mumbai built entirely on open data and open-source tools, addressing a city where choosing a route is, for millions, a health decision.

The system ingests live pollutant readings from all 27 active CPCB and MPCB monitoring stations via the OpenAQ v3 API, converts them to CPCB-compliant National AQI values, and spatially interpolates results across approximately 50,000 road segment midpoints, refreshed every 30 minutes with a historical fallback across 100,464 hourly readings spanning July to December 2025. Four interpolation methods were evaluated through Leave-One-Station-Out cross-validation across 13 temporal slices; IDW ($p=1$) achieved the best overall RMSE of 24.3 AQI units, outperforming Kriging (26.5) and RBF-TPS (42.0). Routes are ranked by Total Dose — the cumulative product of interpolated AQI and segment length — with the cleanest route reducing exposure by 15–40% relative to the fastest at a cost of just 4–12 additional minutes.

Land Use Regression was explored using 101 spatial and temporal predictors, achieving competitive accuracy for NO₂ and SO₂ (RMSE = 10.45 $\mu\text{g}/\text{m}^3$) but near-zero explained variance for PM_{2.5}, establishing a hybrid LUR-interpolation surface as the primary direction for future work. Key limitations include network sparsity across Mumbai's 603 km², a dataset window excluding peak winter months, and a dose model assuming constant breathing rate across transport modes. The system is immediately transferable to any Indian city with CPCB monitoring coverage, with future priorities including wind-driven anisotropic interpolation, a forecasting layer for time-flexible commuters, and a mobile application with personalised dose calculations for vulnerable populations.

1. INTRODUCTION

Air quality has emerged as one of the defining public health crises of the twenty-first century. The accumulation of fine particulate matter, nitrogen dioxide, carbon monoxide, and sulphur dioxide has made breathing outdoors a measurable health risk across rapidly expanding urban centres. In India, 13 of the world's 20 most polluted cities by PM_{2.5} concentration are located in the country, and outdoor air pollution accounts for over 1.67 million deaths annually (WHO, 2023; Lancet, 2020). Mumbai, home to over 20 million people, exemplifies this crisis, combining heavy vehicular traffic, construction, industrial activity, and coastal meteorology into one of South Asia's most complex air quality challenges.

Public infrastructure for communicating air quality information remains fragmented. Adjacent AQI display boards at major junctions have been observed displaying values differing by over 80 units, rooted in inconsistent sensor calibration and data pipelines. No accessible platform currently synthesises readings from Mumbai's 27 CPCB and MPCB monitoring stations into a unified spatial picture. Chronic exposure to PM_{2.5} — frequently exceeding the CPCB's 'Moderate' threshold of 60 $\mu\text{g}/\text{m}^3$ — is associated with asthma, cardiovascular disease, and reduced lung function, disproportionately affecting children, the elderly, and those with pre-existing conditions (CPCB, 2014; WHO, 2021). A shift in health consciousness accelerated by COVID-19 has created genuine demand for tools enabling informed decisions about movement through the city, where choosing a route is a health decision, not merely one of convenience. This project addresses both gaps through a real-time, pollution-aware route planning application for Mumbai. The system ingests live pollutant readings from all active monitoring stations via the OpenAQ v3 API, converts them to a standardised CPCB National AQI score, and uses spatial interpolation to construct a continuous pollution surface across the road network. Users compare routes not only by travel time but by cumulative exposure — total dose, the product of mean AQI and distance — which better represents inhaled pollution than a simple average. Multiple interpolation methods are evaluated using Leave-One-Station-Out cross-validation on 100,464 hourly readings, alongside Land Use Regression as an alternative modelling approach, culminating in a pipeline that transforms sparse monitoring data into an actionable navigation tool.

2. RATIONALE

The motivation for this project arises from three converging realities: the worsening air quality crisis in Mumbai, the inadequacy of existing tools for navigating that crisis, and the growing availability of open-access data that makes a data-driven solution feasible. Mumbai's air quality has deteriorated markedly over the past decade. Construction activity, from infrastructure projects such as the Coastal Road and Metro lines to dense residential development, has added substantial particulate and diesel emissions to an already polluted atmosphere. During the post-monsoon and winter months, PM_{2.5} concentrations routinely breach 150 $\mu\text{g}/\text{m}^3$, placing millions of daily commuters in the 'Very Poor' category of the CPCB scale. Despite this, most residents have no practical means of assessing their personal pollution exposure.

Existing tools fail to bridge this gap. Google Maps optimises routes exclusively for time or distance. Dedicated applications such as IQAir and Plume Labs provide station-level AQI readings but do not integrate with routing. The CPCB's Sameer application displays station readings but offers no spatial interpolation or route planning. No existing tool allows a Mumbai resident to determine which route, at a given hour, minimizes pollution exposure. A compounding problem is the inconsistency of physical AQI infrastructure. Adjacent display boards fewer than five hundred meters apart were directly documented presenting readings differing by more than 80 AQI units simultaneously. This inconsistency, attributable to differing sensor types and averaging periods, renders individual boards untrustworthy and underscores the need for a system that aggregates and standardizes readings across the full monitoring network.

Spatial interpolation was selected as the core modelling approach given the hard constraints imposed by only 27 monitoring stations across Greater Mumbai. Land Use Regression was explored as an alternative but produced near-zero explained variance for PM_{2.5} on held-out stations. Spatial interpolation operates directly on live readings, generalizes robustly with small station counts, and updates automatically, making it well-suited for a real-time application.

This project sits at the intersection of environmental data science, spatial statistics, and public health informatics, demonstrating that open monitoring data combined with validated modelling can produce a tool that is both rigorous and genuinely useful

3. LITERATURE REVIEW

A. Spatial Interpolation of AQI Sensor Data

Pendov, Ji, and Ganchev [1] used live AQI data from an IoT monitoring system to estimate air quality at street-level locations. Their method assigned AQI values to map nodes by interpolating data from the three nearest monitoring stations. This is a useful approach for pollution-aware route planning, but its accuracy depends a lot on how closely the sensors are placed. In areas where monitoring stations are far apart, the estimated AQI values may not represent actual local conditions very well.

In our project, we also focused on using live AQI sensor data, but aimed to improve the interpolation process for Mumbai's sensor distribution. We initially tested Land Use Regression (LUR), but it did not perform well enough without additional complex inputs such as traffic and land-use data. So instead, we used interpolation-based methods like IDW, RBF, and Kriging, and compared them using Leave-One-Station-Out Cross-Validation (LOSO-CV) to choose the most accurate model.

B. Continuous Exposure Mapping and Route Optimization

Grau et al. [2] proposed a pollution-aware pedestrian routing system that uses continuous air quality maps to recommend healthier travel paths. Their study converts atmospheric model outputs into a continuous European Air Quality Index (EAQI) surface and then uses routing algorithms to find paths that balance pollution exposure and travel distance. Similar work by Pendov et al. also considered exposure during travel by combining AQI values with travel time.

However, one limitation of such approaches is that they often depend on model-generated or averaged pollution fields instead of fully real-time observations, which can miss short-term pollution changes. Also, routes chosen only on the basis of lower average AQI may still result in higher total exposure if the route is much longer. In our project, we address this by using only live OpenAQ data and by defining the cleanest route using Total Dose (Mean AQI $\times$ distance in km). This helps ensure that the recommended route is not just cleaner on average, but also practically healthier over the full journey.

4. OBJECTIVES

1. Instantiate live data ingestion pipeline of 27 Mumbai monitoring stations through OpenAQ API, transform pollutant data to CPCB National AQI through official formula, and fall back with historical data.
2. Compare four in the road network of Mumbai: Produce spatially continuous AQI surface through comparing four interpolation methods (IDW, RBF, Kriging) based on Leave-One-Station-Out cross-validation of 100,464 hourly readings.
3. Create route planning that is pollution conscious and ranks alternatives by total dose ($AQI \times \text{distance}$) and route length is a physiologically representative measure of exposure.
4. Evaluate health impacts on the population: Map AQI to CPCB health risk level and examine temporal patterns (hour-by-hour, season, weekday) of pollutant concentrations.
5. Experiment on Land Use Regression with OpenStreetMap characteristics (density of roads, land use, building cover, distance to emission sources) to compare the complementary accuracy to spatial interpolation.
6. Provide interactive web application that allows residents to see live pollution surface, route comparison based on exposure, and opensource methodology that is transferable to other Indian cities.

5. DATA PREPARATION

5.1 Data Source and Extraction

The foundational dataset was constructed from historical air quality data sourced from the OpenAQ platform, which aggregates readings from India's CPCB and Maharashtra Pollution Control Board (MPCB) monitoring networks. Because of the volume of high-frequency sensor data required, readings were extracted programmatically from OpenAQ's public AWS S3 buckets on a day-by-day basis, covering a continuous seven-month window from 1 July 2025 to 31 December 2025. Daily batches were grouped by OpenAQ station ID and chronologically merged into a single consolidated CSV, producing a raw dataset of 118,924 rows across 27 monitoring stations. The period spans both the south-west monsoon (July–September) and the post-monsoon to winter transition (October–December), providing training examples across two climatologically distinct regimes.

The 15 columns fall into three groups: station identifiers (datetime, station_id, station_name, lat, lon), pollutant variables (pm25, pm10, no2, co, so2), and meteorological variables (temp, rh, ws, wd). Pollutant columns represent the five inputs to the CPCB AQI computation; meteorological columns were included to account for atmospheric effects on pollutant dispersion and were used in the LUR modelling component.

Column Name	Description	Category
datetime	Timestamp of the air quality observation	Station/ Temporal
station_id	Unique identifier of the monitoring station	Station Identifier
station_name	Name of the monitoring station	Station Identifier
lat	Latitude of the monitoring station	Spatial
lon	Longitude of the monitoring station	Spatial
source_file	Source file from which the record was merged	Metadata
pm25	PM2.5 concentration (24-hour average, $\mu\text{g}/\text{m}^3$)	Pollutant

Column Name	Description	Category
datetime	Timestamp of the air quality observation	Station/ Temporal
station_id	Unique identifier of the monitoring station	Station Identifier
pm10	PM10 concentration (24-hour average, $\mu\text{g}/\text{m}^3$)	Pollutant
no2	Nitrogen dioxide concentration (1-hour, $\mu\text{g}/\text{m}^3$)	Pollutant
co	Carbon monoxide concentration (8-hour, mg/m^3)	Pollutant
so2	Sulphur dioxide concentration (1-hour, $\mu\text{g}/\text{m}^3$)	Pollutant
temp	Ambient temperature ($^{\circ}\text{C}$)	Meteorological
rh	Relative humidity (%)	Meteorological
ws	Wind speed (m/s)	Meteorological
wd	Wind direction (degrees)	Meteorological

Table 5.1: Dataset Schema

5.2 Preprocessing Pipeline

A structured preprocessing pipeline was applied to address the issues characteristic of multi-station environmental sensor data: duplicates, physically implausible readings, missing values, and outliers.

I. Duplicate removal: Both exact duplicate rows and logical duplicates defined as records sharing the same `station_id` and `datetime` were identified and removed, retaining the first valid occurrence. Rows with a null `station_id` were dropped, as such records cannot be spatially assigned to a monitoring location.

II. Sensor validity checks: Temperature and relative humidity readings of exactly 0.0 were replaced with NaN, since both conditions are climatologically impossible in Mumbai's tropical environment and are characteristic of sensor sentinel errors. Negative pollutant concentrations, which are physically impossible, were similarly nullified across all five pollutant columns.

III. Outlier treatment: Two complementary stages were applied. In the first stage, domain-based hard limits derived from CPCB instrument saturation thresholds were enforced: PM_{2.5} was bounded to 1–600 $\mu\text{g}/\text{m}^3$, PM₁₀ to 1–800 $\mu\text{g}/\text{m}^3$, NO₂ to 0–300 $\mu\text{g}/\text{m}^3$, CO to 0–10 mg/m^3 , and SO₂ to 0–160 $\mu\text{g}/\text{m}^3$. Meteorological variables were constrained to their physically realistic Mumbai ranges (temperature 10–45°C, relative humidity 0–100%, wind speed 0–20 m/s, wind direction 0–360°). Any value outside these bounds was set to NaN. In the second stage, IQR-based Winsorisation was applied per station and per pollutant, with a factor of 3.0 chosen deliberately over the conventional 1.5 to preserve genuine pollution spikes, which are real and recurrent events in Mumbai's urban environment. Values exceeding the station-specific bounds of $Q1 - 3 \times \text{IQR}$ and $Q3 + 3 \times \text{IQR}$ were clipped to the boundary rather than discarded. Side-by-side box plots were generated for each pollutant to confirm the effect of cleaning (Fig 6.1 followed by distribution histograms of the cleaned data with skewness annotations, confirming the expected right-skewed distributions for all particulate variables).

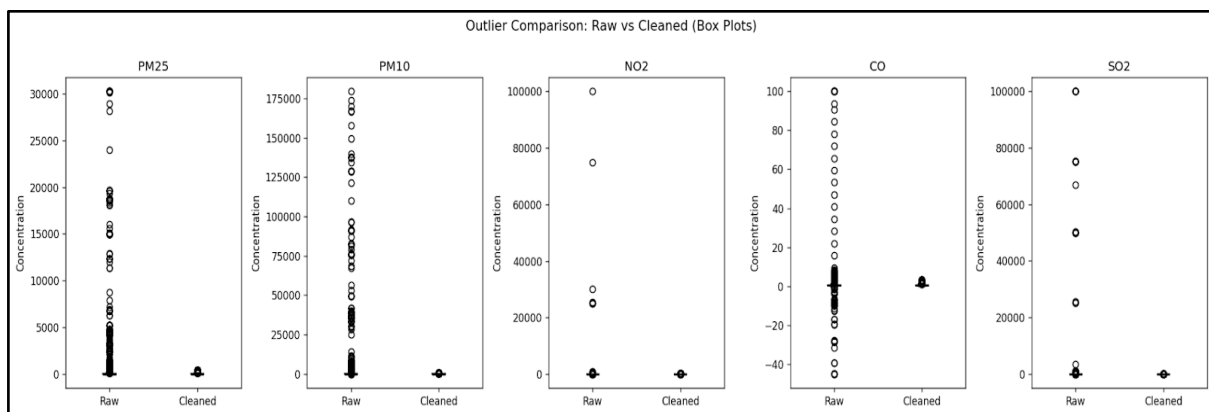


Fig 5.1: Box and Whisker plot for outlier.

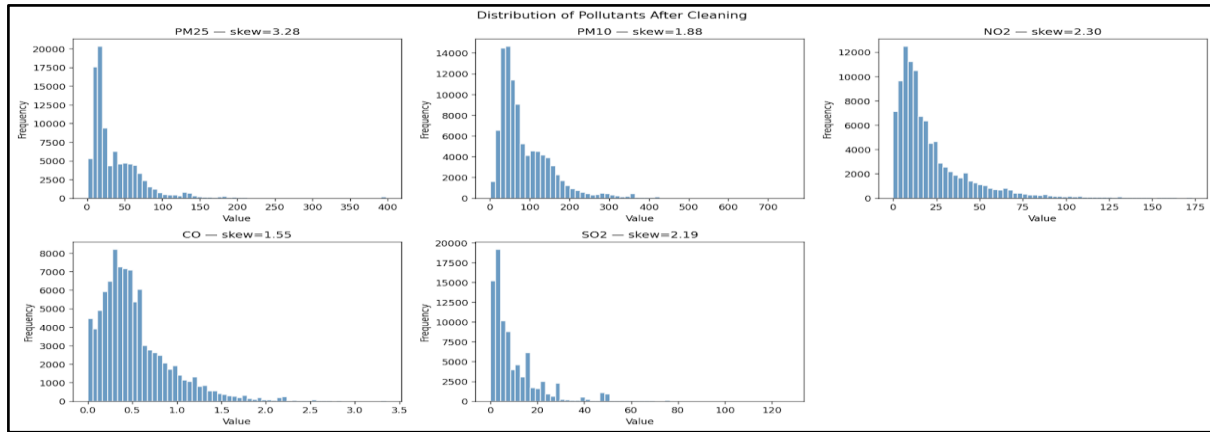


Fig 5.2: Distribution of pollutants after cleaning.

IV. Temporal gap detection: A separate analysis assessed the completeness of the hourly timestamp sequence for each station, independent of whether sensor values were present. For each station, the expected set of hourly timestamps was constructed from its first to last recorded datetime and compared against the actual timestamps recorded. Temporal coverage was expressed as a percentage and stations were colour-coded into three tiers: below 80% (flagged red), 80–95% (blue), and 95% or above (green). This identified stations with significant structural outages and informed decisions about interpolation thresholds.

V. Missing value imputation: A conservative, three-phase imputation strategy was applied. Prior to imputation, a consecutive gap analysis identified the maximum run of consecutive NaN values per station and column, establishing safe interpolation limits — gaps of 48–72 hours cannot responsibly be filled from adjacent readings. In the first phase, gaps of up to three consecutive hours were filled using time-weighted linear interpolation (`interpolate(method='time', limit=3)`), which weights imputed values by actual elapsed time rather than treating all steps as equal. Longer gaps were left as NaN. The second phase addressed the higher missingness in temp and rh — attributable to limited anemometer and thermometer deployment — using a distance-weighted KNN imputer ($k=5$) operating across stations in pivot-table format, so that cross-station temporal correlations informed the fill. Residual missingness was resolved through a cascading climatological fallback: values were replaced first with the station–month–hour mean, then the station–hour mean across all months, and finally the cross-station hourly mean. This hierarchy preserved diurnal and seasonal patterns as precisely as the data allowed, degrading gracefully to broader estimates when finer-grained group means were unavailable.

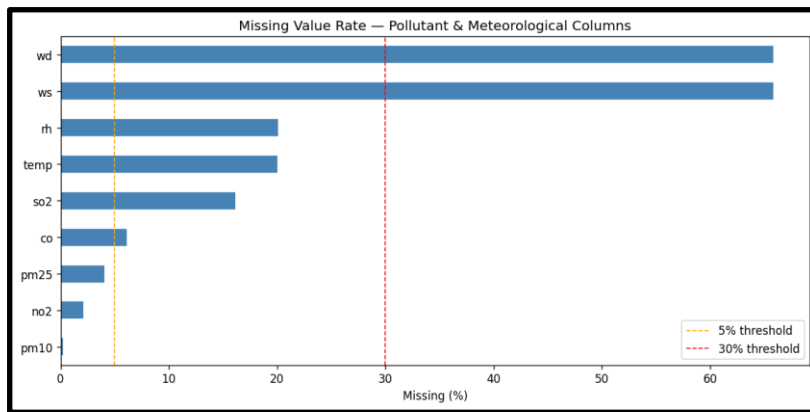


Fig 5.3: Bar Graph of missing values.

5.3 AQI Computation

Following preprocessing, the Indian AQI was computed for each observation using the CPCB sub-index method. For each pollutant, a sub-index was derived by piecewise linear interpolation over the official CPCB breakpoint table:

$$I = \frac{I_{hi} - I_{lo}}{C_{hi} - C_{lo}}(C - C_{lo}) + I_{lo}$$

where C is the observed concentration and the bounds correspond to the applicable breakpoint interval. Sub-indices exceeding the highest defined breakpoint were capped at 500. The station AQI was taken as the maximum sub-index across all available pollutants, consistent with the CPCB reporting standard. A minimum of three valid sub-indices was required for a composite AQI to be considered reliable; rows with fewer valid pollutants were assigned NaN and excluded from surface interpolation. Each observation was additionally assigned a CPCB category label: Good (0–50), Satisfactory (51–100), Moderate (101–200), Poor (201–300), Very Poor (301–400), Severe (401–500) stored alongside the numeric AQI.

5.4 Correlation Analysis

A Pearson correlation matrix was computed across all five pollutants, four meteorological variables, and the computed AQI, and visualised as a lower-triangular heatmap. The strongest relationships were between PM2.5 and PM10 ($r = 0.81$) and between both particulate variables and composite AQI ($r = 0.91$), confirming that particulate matter dominates the AQI signal in this dataset. The moderate correlation between NO₂ and CO ($r = 0.40$) is consistent with a shared traffic emission source. Temperature and relative humidity showed weak negative correlations with most pollutants, reflecting their role in atmospheric dispersion rather than emission. Wind speed and direction showed near-zero correlations with pollutant

concentrations, likely a consequence of the high missingness in those columns reducing their statistical signal. This analysis informed feature selection for the LUR modelling stage and provided interpretive context for how meteorological conditions interact with the pollution field across Mumbai's monitoring network.

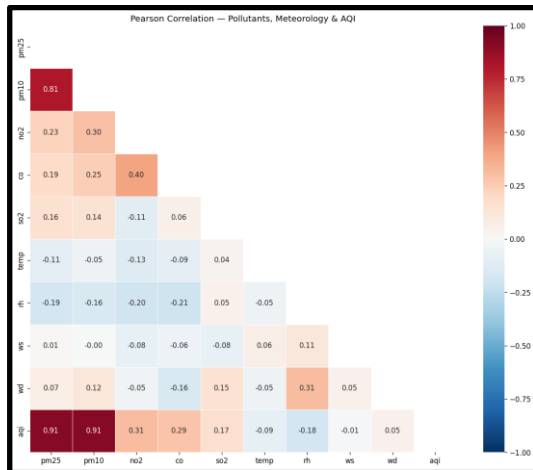


Fig 5.4 Correlation matrix of attribute

6. METHODOLOGY

6.1 System Architecture and End-to-End Pipeline

The system operates as a two-phase pipeline. In the Data Phase, live pollutant concentrations are fetched from the OpenAQ v3 API, converted to CPCB-compliant AQI values per station, and spatially interpolated across Mumbai's road network to produce a continuous pollution surface. In the Routing Phase, the user's origin and destination are geocoded to coordinates, multiple route alternatives are fetched from OpenRouteService, and each route's pollution exposure is computed by sampling the surface along the full route geometry.

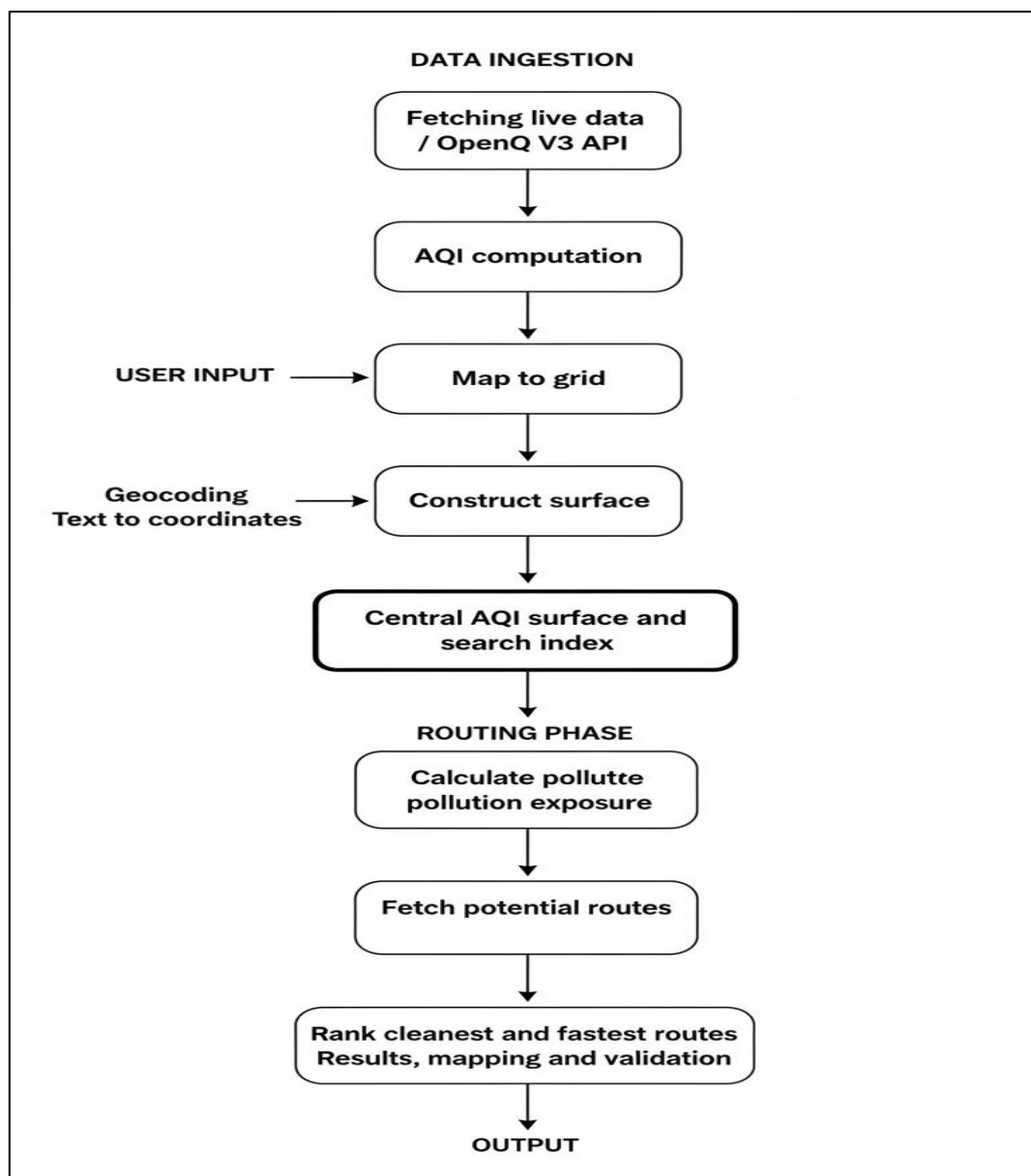


Fig 6.1: Block Diagram of the Mumbai AQI Real-Time Route Planner.

6.2 CPCB Air Quality Index Computation

All AQI values in this project are computed according to the National Air Quality Index standard published by India's Central Pollution Control Board (CPCB, 2014), not the US EPA standard used by most global air quality applications. The two standards use different pollutant averaging periods, different breakpoint tables, and produce different numeric values from the same raw concentration data. Using the CPCB standard ensures that values produced by this system are directly comparable to those published on India's official national AQI dashboard. The CPCB AQI is a composite index computed from five pollutants: PM_{2.5} (24-hour average), PM₁₀ (24-hour average), NO₂ (1-hour average), CO (8-hour average), and SO₂ (1-hour average). The index runs from 0 to 500, divided into six health categories: Good (0–50), Satisfactory (51–100), Moderate (101–200), Poor (201–300), Very Poor (301–400), and Severe (401–500).

6.2.1 Sub-Index Calculation

For each pollutant, a sub-index is computed by applying piecewise linear interpolation over the CPCB breakpoint table. Within any concentration interval $[C_{lo}, C_{hi}]$ mapped to AQI interval $[I_{lo}, I_{hi}]$, the sub-index scales linearly with concentration:

$$I = \frac{I_{hi} - I_{lo}}{C_{hi} - C_{lo}}(C - C_{lo}) + I_{lo}$$

where C is the measured pollutant concentration. If C exceeds the highest defined breakpoint, its sub-index is capped at 500.

PM _{2.5} Conc. ($\mu\text{g}/\text{m}^3$, 24h avg)	AQI Sub-Index Range	Category	Health Impact
0 – 30	0 – 50	Good	Minimal impact
30 – 60	51 – 100	Satisfactory	Minor breathing discomfort for sensitive persons
60 – 90	101 – 200	Moderate	Breathing discomfort for asthma, lung and heart disease patients

90 – 120	201 – 300	Poor	Breathing discomfort for most on prolonged exposure
120 – 250	301 – 400	Very Poor	Respiratory illness on prolonged exposure
250 – 380	401 – 500	Severe	Serious risk for healthy people; hazardous for sensitive groups

Table 6.1: CPCB PM_{2.5} breakpoint table used for sub-index computation (CPCB, 2014).

6.2.2 Composite Station AQI

Once a sub-index has been computed for each available pollutant, the station's overall AQI is the maximum sub-index across all pollutants:

$$AQI = \max\{I_{PM2.5}, I_{PM10}, I_{NO_2}, I_{CO}, I_{SO_2}\}$$

The pollutant with the highest sub-index is designated the dominant pollutant. This maximum-based aggregation ensures the AQI reflects the worst health risk present at the station. A minimum of three valid pollutant sub-indices is required for a composite AQI to be considered reliable; observations with fewer are excluded from surface interpolation.

6.2.3 Unit Harmonization

Raw readings from the OpenAQ API require unit harmonisation before AQI computation. The CPCB breakpoint table specifies CO in mg/m³, but several sensors report CO in µg/m³. Since urban CO never exceeds ~20 mg/m³ under severe conditions, any CO reading exceeding 50 is automatically converted by dividing by 1,000. Physically implausible readings (PM_{2.5} > 1,000 µg/m³, PM₁₀ > 1,500 µg/m³, NO₂ > 2,000 µg/m³) are rejected as sensor errors before AQI computation.

6.3 Live Data Ingestion via OpenAQ v3 API

Live pollutant readings are fetched from the OpenAQ v3 API, which aggregates data from India's CPCB and Maharashtra Pollution Control Board (MPCB) monitoring networks. The v3 API requires a two-step interaction pattern.

6.3.1 Two-Step Sensor Resolution

In the first step, a GET request is issued to `/v3/locations` with a bounding box covering the Mumbai metropolitan area (18.89°N–19.32°N, 72.79°E–72.96°E). This returns all active monitoring stations with their sensor lists. Each sensor has a numeric ID and a parameter name (e.g. 'pm2.5', 'no2'). A global sensor map is built from this response:

$$\text{sensor_map} : \{ \text{sensor_id} \rightarrow \text{pollutant_code} \}$$

In the second step, time-averaged concentrations are fetched from `/v3/sensors/{id}/hours` for each sensor, using the CPCB-mandated averaging windows: 24 hours for PM2.5 and PM10, 8 hours for CO, 1 hour for NO₂ and SO₂. If hourly history is insufficient for a station, the system falls back to `/v3/locations/{id}/latest` for an instantaneous snapshot, which is flagged in the interface.

6.3.2 Parallel Fetching and Caching

With 27 stations, sequential API calls would take approximately 40 seconds. The system uses Python's `ThreadPoolExecutor` with up to 10 concurrent worker threads, reducing total fetch time to 3–4 seconds. A UTC timestamp of the most recent successful fetch is stored in Streamlit session state. A fresh fetch is triggered only when the cache age exceeds 30 minutes, or when the user manually refreshes, or when the interpolation method changes since a different method applied to the same station data produces a different spatial surface.

6.3.3 Historical Fallback

If fewer than five stations return valid live readings, the system falls back to a pre-computed historical surface derived from the cleaned dataset (July–December 2025). The current month is mapped to the closest available historical month based on Mumbai's climate zones, and the closest matching hour of day and weekday/weekend type is selected, ensuring the application remains informative even without live data.

6.4 Coordinate Reference Systems

All raw data arrives in WGS84 (EPSG:4326), where positions are expressed as decimal-degree latitude and longitude. WGS84 is unsuitable for metric computations because degrees are not uniform in metres; one degree of longitude covers 98 km at Mumbai's latitude of 19°N,

compared to 111 km at the equator. All spatial operations (buffer creation, interpolation grid construction, KD-tree indexing, distance calculations) are therefore performed in UTM Zone 43N (EPSG:32643), the metric Cartesian system for central India, where both axes are in metres and distances are accurate to within 0.04% across Mumbai's extent. Station coordinates are projected to UTM immediately after ingestion.

6.5 Spatial Interpolation of the AQI Surface

The 27 monitoring stations provide AQI values at 27 discrete locations. To estimate AQI at the approximately 50,000 road segment midpoints across Mumbai's road network, a continuous pollution surface must be constructed from these sparse observations. All interpolation is performed in UTM coordinates. A regular grid of 200-metre resolution is laid over Mumbai's bounding box, producing approximately 25,000 grid points. The chosen interpolation method is applied at these points to generate a continuous 2D surface, which is then sampled directly at road segment midpoint coordinates.

6.5.1 Inverse Distance Weighting (IDW)

Inverse Distance Weighting is the most widely used deterministic spatial interpolation method. Its fundamental assumption is Tobler's First Law of Geography: nearby locations have more similar values than distant ones. IDW assigns every query point a weighted average of all station values, where the weight of each station decays as a power function of its distance from the query point. The weight of station i at query point x is:

$$w_i(x) = \frac{1}{(d_i(x))^p + \epsilon}$$

where $d_i(x)$ is the Euclidean UTM distance in metres from x to station i , p is the power parameter, and $\epsilon = 10^{-10}$ prevents division by zero when a query point coincides exactly with a station. The interpolated AQI at x is:

$$Z(x) = \frac{\sum_{i=1}^n w_i(x) Z_i}{\sum_{i=1}^n w_i(x)}$$

where Z_i is the AQI at station i . Two values of p are implemented. With $p = 1$, weights decay linearly with distance, producing a smoother surface where distant stations retain meaningful influence. With $p = 2$ (the standard choice for environmental applications), weights decay quadratically, halving the distance quadruples the weight producing a more localised surface. Higher values of p produce sharp 'bullseye' artefacts around each station

and are not implemented. IDW is fast, always converges, and requires no external dependencies, making it the recommended default for the live application.

6.5.2 Radial Basis Function Interpolation (RBF Thin Plate Spline)

RBF interpolation constructs a globally smooth surface by expressing the interpolated field as a linear combination of radially symmetric basis functions centred at each station:

$$Z(x) = \sum_{i=1}^n \lambda_i \phi(\|x - x_i\|)$$

where x_i is the UTM coordinate of station i , $\|x - x_i\|$ is the Euclidean distance from the query point to station i , ϕ is the kernel function, and λ_i are weights determined by solving a linear system. The weights are found by requiring the surface to pass through all station values simultaneously:

$$\Phi \lambda = Z_{\text{obs}}$$

where Φ is the $n \times n$ matrix of kernel values between all station pairs and Z_{obs} is the vector of observed AQI values. This project uses the thin plate spline kernel:

$$\phi(r) = r^2 \ln(r)$$

The thin plate spline minimises the surface's bending energy, the integral of squared second-order spatial derivatives producing the smoothest possible surface consistent with the data. This is physically well-suited to urban air quality fields, which vary continuously without sharp discontinuities. Before fitting, all UTM coordinates are normalised by dividing by their standard deviation to prevent numerical instability from large coordinate magnitudes ($\sim 10^6$ m).

6.5.3 Ordinary Kriging

Ordinary Kriging is the geostatistically optimal interpolator, a Best Linear Unbiased Estimator (BLUE) that minimises prediction variance while ensuring predictions are unbiased. Unlike IDW and RBF, Kriging uses the empirical spatial correlation structure of the data, quantified through the variogram.

The experimental variogram $\gamma(h)$ measures how dissimilarity between observations increases with separation distance h :

$$\gamma(h) = \frac{1}{2N(h)} \sum_{i=1}^{N(h)} [Z(x_i) - Z(x_i + h)]^2$$

where $N(h)$ is the number of station pairs separated by distance approximately h . A theoretical spherical variogram model is fitted to this empirical curve:

$$\gamma(h) = \begin{cases} c_0 + c \left[\frac{3}{2} \left(\frac{h}{a} \right) - \frac{1}{2} \left(\frac{h}{a} \right)^3 \right], & h \leq a \\ c_0 + c, & h > a \end{cases}$$

where c_0 is the nugget (variance at $h = 0$, representing measurement error), c is the partial sill (spatial variance), and a is the range (distance at which correlation drops to zero). The spherical model is standard for environmental variables: it assumes correlation reaches zero at a finite range, which is physically reasonable for urban pollution.

For each prediction point x , optimal weights are found by solving the augmented Kriging system:

$$\mathbf{K} \boldsymbol{\lambda} = \mathbf{k}$$

where $\mathbf{K}$ is the $(n+1) \times (n+1)$ matrix of variogram values between station pairs (augmented with a row and column of ones for the Lagrange multiplier enforcing $\sum \lambda_i = 1$), and $\mathbf{k}$ is the corresponding vector of variogram values between stations and the prediction point. The prediction is:

$$Z^*(x) = \sum_{i=1}^n \lambda_i Z(x_i)$$

Kriging also produces the prediction variance $\sigma^2_K(x)$ as a by-product, providing spatially varying uncertainty estimates not available from IDW or RBF. It is implemented using the pykrige library; if unavailable, the system falls back to IDW ($p=2$).

6.5.4 Method Comparison

Method	Key Properties	Strengths	Limitations
IDW ($p=2$)	Quadratic distance-decay, exact interpolation	Fast, simple, always converges, no dependencies	Bullseye artefacts, no uncertainty estimate
IDW ($p=1$)	Linear distance-decay, smoother surface	Reduced artefacts, best overall LOSO RMSE	May under-weight local gradients near sources
RBF – TPS	Minimizes bending energy, global smooth fit	No bullseye artefacts, physically smooth	Sensitive to outlier stations, no uncertainty

Ordinary Kriging	BLUE estimator, variogram-based spatial structure	Statistically optimal, provides prediction variance	Requires enough stations for variogram fit; slower
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Table 6.2: Comparison of the four interpolation methods implemented in this project.

6.6 Interpolation Evaluation: Leave-One-Station-Out Cross-Validation

To determine which interpolation method produces the most accurate pollution surface under standard, reproducible conditions, all four methods were evaluated using Leave-One-Station-Out Cross-Validation (LOSO CV) conducted on the historical dataset of 100,464 hourly readings spanning July to December 2025. Historical data was deliberately chosen for this evaluation rather than live data: it provides a stable, fixed benchmark where true station values are known with certainty, enabling a fair and repeatable comparison across all methods under identical conditions. The results are pre-computed and stored in `loso_cv_results.json`, which the application loads at startup to populate the Interpolation Evaluation tab.

6.6.1 LOSO CV Algorithm

LOSO CV is a special case of k-fold cross-validation where k equals the number of stations ($n = 27$). For each station i :

1. Remove station i from the dataset, leaving $n-1 = 26$ training stations.
2. Fit the interpolation method using the 26 remaining stations and their AQI values.
3. Predict the AQI at station i 's UTM coordinates using only the fitted surface.
4. Record the signed error: $e_i = Z^*_i - Z_i$
5. Repeat for all 27 stations and aggregate errors into performance metrics.

This procedure is maximally data-efficient and provides an honest estimate of how well each method generalizes to an unmonitored location which is precisely the task the system performs when estimating AQI at road segments between stations. LOSO CV was run across 13 temporal slices: Overall, six individual months (July–December), four time-of-day windows (Night 0–6h, Morning rush 7–10h, Midday 11–16h, Evening rush 17–22h), and Weekday versus Weekend.

6.6.2 Performance Metrics

Four metrics are computed from the 27 signed errors $\{e_1, \dots, e_n\}$:

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n e_i^2}, \quad \text{MAE} = \frac{1}{n} \sum_{i=1}^n |e_i|, \quad \text{Bias} = \frac{1}{n} \sum_{i=1}^n e_i, \quad R^2 = 1 - \frac{\sum_{i=1}^n e_i^2}{\sum_{i=1}^n (Z_i - \bar{Z})^2}$$

RMSE (Eq. 14) penalises large errors heavily due to squaring and is the primary model selection criterion, with values below 30 AQI units considered good for this application. MAE (Eq. 15) treats all errors equally and is more robust to outliers. Bias (Eq. 16) is the signed mean error: positive values indicate systematic over-prediction, negative values indicate under-prediction, and values near zero are ideal. R^2 (Eq. 17) expresses the proportion of AQI variance explained; values below zero indicate the method performs worse than simply predicting the global mean.

6.7 Pollution-Aware Route Planning

The route planning component takes a user-specified origin and destination, fetches multiple route alternatives from OpenRouteService, and evaluates each route's pollution exposure using the AQI surface from Section 7.5. The result is a ranked set of routes presented with full exposure metrics, a trade-off summary, and a segment-level AQI profile chart.

6.7.1 Geocoding

User-entered location names are converted to geographic coordinates using Photon, a geocoding API built on OpenStreetMap data, biased toward the Mumbai metropolitan area. A three-attempt fallback strategy is employed: first, the query is submitted with ', Mumbai, India' appended and a viewbox constraint bounded to Mumbai's bounding box; if this fails, the viewbox is relaxed; and if this also fails, the raw query is submitted globally. Results are cached using Python's `lru_cache` decorator to avoid redundant calls during Streamlit's per-interaction script re-execution.

6.7.2 Route Computation: Dijkstra's Algorithm and Contraction Hierarchies

Routes are computed by OpenRouteService (ORS), an open-source routing engine built on the OpenStreetMap road graph. The fundamental algorithm underlying all road routing is Dijkstra's shortest path algorithm, which finds the minimum-cost path between two nodes in a weighted graph.

In Dijkstra's algorithm, every road intersection is a node and every road segment is a directed edge with a weight equal to its traversal cost (travel time for the fastest route, or distance for the shortest). The algorithm maintains a priority queue of candidate nodes ordered by their

cumulative cost from the origin. At each step, the node with the lowest cumulative cost is expanded: for each of its outgoing edges, if the path through this node improves the currently known cost to the neighbour, the neighbour's cost is updated. This process continues until the destination node is settled:

$$\text{cost}(v) = \min_{\text{paths } P \text{ from } s \text{ to } v} \sum_{e \in P} w(e)$$

where s is the source node, v is any node, and $w(\text{edge})$ is the edge weight (travel time in seconds for the fastest route). Dijkstra's algorithm guarantees finding the globally optimal path when all edge weights are non-negative, which holds for travel-time routing.

For a city-scale road graph like Mumbai's OSM network (which contains hundreds of thousands of nodes and edges), vanilla Dijkstra is computationally expensive. ORS therefore uses Contraction Hierarchies (CH), a preprocessing-based acceleration technique. During preprocessing, less important nodes are iteratively 'contracted' out of the graph, and shortcut edges are added to preserve shortest-path distances. At query time, a bidirectional Dijkstra search is run upward through the hierarchy from both origin and destination simultaneously, meeting in the middle. This reduces query time from seconds to milliseconds, making real-time routing feasible.

For the fastest route, the edge weight is the travel time in seconds, derived from the road's speed limit and length. For alternative routes, ORS varies the weighting to explore paths that differ meaningfully from the fastest option, controlled by two parameters: `weight_factor` = 1.6 (alternatives may be up to 60% longer in travel cost) and `share_factor` = 0.6 (alternatives must differ in at least 40% of their geometry from the fastest route). Up to three routes are returned per request.

6.7.3 KD-Tree Road Segment Snapping

Each route is returned as a sequence of [longitude, latitude] waypoints. Matching each waypoint to its nearest road segment in the AQI surface dataset requires a nearest-neighbour search across approximately 50,000 road midpoints. A brute-force search would require 50,000 distance comparisons per waypoint. A KD-tree (K-Dimensional tree) reduces this to $O(\log N)$ per query.

The KD-tree partitions the 2D coordinate space by recursively splitting points along alternating axes, choosing the median at each split to produce a balanced binary tree. For a nearest-neighbour query at point P , the tree is traversed to the leaf cell containing P ; the algorithm then backtracks, pruning branches whose bounding hyperrectangle is further from P than the current

best candidate. With $N = 50,000$ road midpoints, this requires approximately $\log_2(50,000) \approx 16$ comparisons rather than 50,000. A snapping threshold of 500 metres is applied: waypoints whose nearest road midpoint is further than 500 m (e.g. over water) are excluded from exposure computation.

6.7.4 Haversine Segment Lengths

The physical length of each segment between consecutive waypoints is computed using the Haversine formula for great-circle distance:

$$a = \sin^2\left(\frac{\Delta\phi}{2}\right) + \cos(\phi_1) \cos(\phi_2) \sin^2\left(\frac{\Delta\lambda}{2}\right), \quad d = 2R \cdot \text{atan2}(\sqrt{a}, \sqrt{1-a}), \quad R = 6,371,000 \text{ m}$$

where ϕ_1, ϕ_2 are endpoint latitudes in radians, $\Delta\phi$ and $\Delta\lambda$ are latitude and longitude differences in radians, and d is the segment length in metres. The atan2 form is used for numerical stability. The formula is accurate to within 0.3% for intra-city distances of 2–30 km.

6.7.5 Pollution Exposure Metrics

Two exposure metrics are computed for each route. For each consecutive waypoint pair i in the route geometry, let AQI_i denote the AQI of the nearest road segment (from the interpolated surface) and L_i the Haversine length in metres.

The distance-weighted mean AQI is:

$$\text{Mean AQI} = \frac{\sum_{i=1}^n AQI_i L_i}{\sum_{i=1}^n L_i}$$

This weights each segment's AQI contribution by its physical length, so long stretches through polluted roads contribute proportionally more than short stretches through clean air. However, mean AQI alone does not account for route length: a 10 km route through moderate pollution may be more harmful than a 2 km route through heavy pollution.

Total dose is defined explicitly as the cumulative exposure over the entire journey:

$$\text{Total Dose} = \sum_{i=1}^n AQI_i L_i \quad [\text{AQI} \cdot \text{m}]$$

Total dose is directly proportional to the total volume of polluted air inhaled, under the assumption of constant breathing rate and tidal volume throughout the journey. This is the primary criterion for selecting the cleanest route, as it captures both the intensity and duration of pollution exposure. The cleanest route is therefore:

$$\text{Cleanest Route} = \arg \min_r \{\text{Total Dose}(r)\}$$

This formulation is an original design decision of this project. It is motivated by the established health principle that cumulative pollutant intake, not instantaneous peak concentration, drives the primary respiratory and cardiovascular harm from urban travel. A user passing briefly through a high-AQI zone accumulates far less total exposure than a user travelling a long route through even moderately polluted roads.

6.7.6 Route Ranking and Trade-off Quantification

After exposure metrics are computed for all returned routes, two routes are tagged. The fastest route is:

$$\text{Fastest Route} = \arg \min_r \{\text{duration}_s(r)\}$$

where duration_s is the travel time in seconds as reported by ORS. When the fastest and cleanest routes are the same, this is explicitly flagged in the interface.

When the fastest and cleanest routes differ, a trade-off summary is computed and displayed to the user. Let F denote the fastest route and C the cleanest route. The trade-off quantities are:

These three quantities together tell the user: how much extra time the cleaner route costs (ΔTime), how much total pollution exposure is avoided (ΔDose), and how much lower the average AQI is along the cleaner route (ΔAQI). The application presents all three values in plain language alongside the route comparison cards, enabling the user to make an informed trade-off decision based on their own priorities.

For each selected route, a segment-level AQI profile chart is displayed, a bar chart in which each bar represents one road segment, coloured according to its CPCB AQI category, and a dashed horizontal line marks the route's mean AQI. This allows the user to see not just the aggregate exposure metric but the spatial distribution of pollution along the route from origin to destination.

6.8 Streamlit Web Application

The entire pipeline described in the preceding sections is delivered as an interactive web application built with Streamlit, a Python framework that converts Python scripts into browser-based UIs without requiring front-end code. The application is structured as a single-page interface with a persistent sidebar and four tabbed panels, each corresponding to a major functional module of the system.

6.8.1 Sidebar Configuration and Data Status

The sidebar provides the application's global configuration layer. Users enter their OpenAQ API key (for live data access) and their OpenRouteService API key (for routing). A dropdown selector allows the user to choose the interpolation method, which triggers a surface rebuild if the selection changes. A data status panel displays whether the application is running on live data or the historical fallback, the age of the current cache in minutes, how many stations used the CPCB-correct time-averaged readings versus the instantaneous fallback, and which interpolation method was used to build the current surface. A Refresh AQI Data button manually invalidates the cache and forces a new fetch.

6.8.2 Tab 1: Interpolation Evaluation

The first tab presents the pre-computed LOSO CV results from `loso_cv_results.json`. A group selector allows the user to explore results across three slice categories: Overall and day-type (Weekday/Weekend), Monthly (July–December), and Time of Day (four windows). Within the selected group, a slice selector drills into a specific time slice. Results are displayed as a formatted metrics table with the best-performing method highlighted per slice followed by three bar charts (RMSE, MAE, Bias) for the selected slice and a multi-line RMSE trend chart showing how each method's error varies across all 13 temporal slices simultaneously. A grouped RMSE vs MAE chart for the Overall slice rounds out the comparison. A recommendation banner at the top of the tab always identifies the best method across all slices based on overall RMSE.

6.8.3 Tab 2: AQI Surface Explorer

The second tab displays the current pollution surface as an interactive Folium map. Road segment midpoints are rendered as colour-coded circular markers following the CPCB palette (green for Good through dark red for Severe), sampled to a maximum of 12,000 markers for browser rendering performance. Monitoring station markers are overlaid, coloured by their live AQI value if live data is active. Summary statistics city mean AQI, median AQI, maximum AQI, total road segments, and current method are displayed as metric cards above the map. A bar chart of road segments by CPCB category quantifies the distribution of pollution levels across the road network. The map and statistics update automatically when the interpolation method is changed.

6.8.4 Tab 3: Route Planner

The third and primary tab provides the pollution-aware navigation interface. Two search widgets, one for origin, one for destination, use the Photon geocoding API to provide live location suggestions as the user types, filtered to the Mumbai metropolitan area. A transport mode selector offers three options: Car, Walking, and Cycling, corresponding to the ORS profiles driving-car, foot-walking, and cycling-regular respectively. The Find Routes button is enabled only when both locations are selected and an ORS API key is present.

Upon clicking Find Routes, the application fetches up to three route alternatives, computes pollution exposure for each, tags the fastest and cleanest, and displays the results. The fastest and cleanest routes are presented as side-by-side comparison cards showing travel time, distance, mean AQI with CPCB category badge, and total dose. When the two routes differ, a trade-off summary (Equations 25–27) is displayed in plain language. An interactive Folium route map renders all routes as polylines: blue for fastest, green for cleanest, grey for alternatives with road-level AQI dots and origin/destination markers. A segment-level AQI profile bar chart is displayed for the currently selected route.

6.8.5 Tab 4: LUR Model Comparison

The fourth tab documents the Land Use Regression modelling effort as a methodological comparison. It presents the LUR approach against spatial interpolation in a structured side-by-side description, the hold-out performance metrics for all five models across all five pollutants as bar charts (RMSE and MAE per pollutant), a cross-method comparison table juxtaposing the best LUR RMSE against the best interpolation RMSE, and a key takeaways section discussing the relative strengths, limitations, and the rationale for retaining interpolation as the primary method. This tab serves both as a research record and as a demonstration that the team rigorously explored and compared alternative modelling approaches.

6.8.6 Session State and Reactivity

Streamlit re-executes the entire Python script on every user interaction. To prevent redundant API calls, surface rebuilds, and route recomputation, all expensive outputs are stored in Streamlit's session_state dictionary: the live stations DataFrame, road AQI surface DataFrame, KD-tree, last fetch timestamp, current surface method, computed routes, and selected route index. Each component checks whether its inputs have changed before recomputing. This architecture ensures that changing the selected route on the map, for example, does not trigger a new API fetch or surface rebuild, while changing the interpolation method does.

6.9 Land Use Regression (LUR) Exploratory Modelling

In parallel with the spatial interpolation approach, this project explored Land Use Regression (LUR) as an alternative methodology for estimating pollutant concentrations across the Mumbai road network. LUR is a well-established technique in environmental epidemiology. Unlike spatial interpolation, which estimates values at unmonitored locations by weighting known station readings, LUR learns the statistical relationship between a location's environmental context its surrounding road network, land use, building density, vegetation, and proximity to emission sources and the pollutant concentrations observed there. The LUR models were not deployed in the live application; spatial interpolation was retained as the primary method for reasons discussed in Section 5.9.6.

6.9.1 Feature Engineering

Spatial predictor variables were computed for each of the 27 monitoring stations from OpenStreetMap data using OSMnx, projected to UTM Zone 43N. Since station locations are fixed, spatial features are constants at the station level, computed once and broadcast to all hourly observations via `station_id` join.

Road network features (42 features) were computed at four buffer radii (100 m, 300 m, 500 m, 1,000 m): total road length, road density (m/m^2), road length by tier (major/secondary/local), intersection count, and intersection density. Distance to the nearest major and secondary road were computed as scalar features. Point source distances (5 features) captured proximity to petrol stations, bus terminals, railway stations, and airport infrastructure. Land use proportions (27 features) measured industrial, commercial, residential, and green area fractions within 300 m, 500 m, and 1,000 m buffers. Vegetation and water features (10 features) included green area proportions at 100 m, 300 m, and 500 m buffers, distance to the nearest water body, and water coverage within 500 m. Building footprint density (11 features) captured building area, count, and coverage ratio at 500 m and 1,000 m buffers. Elevation (1 feature) was obtained from the Open-Elevation API.

Temporal features were derived row-by-row from the datetime column. Cyclic sine/cosine encodings were applied to hour, day of week, and month to preserve their circular nature. Binary flags for `is_weekend` and `is_rush_hour` (07:00–10:00 and 17:00–20:00 IST), a season code (Monsoon=0, Post-Monsoon=1, Winter=2), and a continuous time-of-day variable were also included. After merging all feature blocks with the cleaned hourly time series by

station_id, the final dataset contained 100,464 rows and 124 columns, of which approximately 101 served as predictors.

6.9.2 Target Variables and Train-Test Split

Separate LUR models were trained for five pollutant targets: PM_{2.5}, PM₁₀, NO₂, CO, and SO₂. The train-test split was conducted at the station level: entire stations were assigned to either training (~80%, approximately 21–22 stations) or test (~20%, approximately 5–6 stations), ensuring the model is evaluated on spatially unseen locations.

Target	Total Rows	Train Rows	Test Rows	Units
PM _{2.5}	94,534	76,467	18,067	µg/m ³
PM ₁₀	97,852	77,630	20,222	µg/m ³
NO ₂	99,212	79,697	19,515	µg/m ³
CO	94,797	75,931	18,866	mg/m ³
SO ₂	90,212	72,086	18,126	µg/m ³

Table 6.3: Station-wise train-test split sizes per pollutant target.

6.9.3 Models and Spatial Cross-Validation

Five regression algorithms were compared: Ridge Regression and ElasticNet as linear baselines, and Random Forest, Gradient Boosting, and XGBoost as non-linear ensemble methods. All models were embedded in a KNN imputation (k=5) → RobustScaler preprocessing pipeline. Random Forest and XGBoost were further tuned using RandomizedSearchCV (30 iterations) with 5-fold GroupKFold cross-validation grouped by station ID ensuring validation folds always contain stations not seen during training within that fold, preventing inflated performance estimates from temporal autocorrelation within stations.

6.9.4 LUR Results

Pollutant	Best Model	Hold-out RMSE	Hold-out MAE	Units
PM _{2.5}	ElasticNet	24.05	16.97	µg/m ³
PM ₁₀	XGBoost	41.89	32.45	µg/m ³
NO ₂	Random Forest	10.45	7.90	µg/m ³
CO	ElasticNet	0.42	0.29	mg/m ³
SO ₂	Random Forest	10.45	6.81	µg/m ³

Table 6.4: Best LUR model and hold-out performance per pollutant target.

NO₂ was the most successfully modelled pollutant (RMSE = 10.45 µg/m³), consistent with the LUR literature: NO₂ has a strong and spatially structured relationship with road proximity and traffic density, which are well represented in the feature set. SO₂ achieved a comparable RMSE of 10.45 µg/m³. PM₁₀ modelling with XGBoost yielded RMSE = 41.89 µg/m³, with ensemble models consistently outperforming linear ones, suggesting meaningful non-linear relationships between particulate matter and spatial predictors.

PM_{2.5} modelling was the most challenging. While ElasticNet achieved the lowest hold-out RMSE of 24.05 µg/m³, this corresponds to near-zero or negative R² on held-out stations, indicating the model explains little of the cross-station variance. This is not unexpected: PM_{2.5} in Mumbai is strongly driven by mesoscale meteorological dynamics – the Arabian Sea sea breeze during monsoon, temperature inversions during winter – which static land-use predictors cannot capture. CO modelling was similarly weak, reflecting CO's dependence on highly localised, short-duration emission events.

6.9.5 Limitations of the LUR Approach

Several structural limitations constrained LUR performance in this project. The dataset covered only six months (July–December 2025), limiting the model's ability to learn across the full annual cycle of Mumbai's monsoon–post-monsoon–winter climate regimes. With only 27 monitoring stations, the effective number of spatially independent training locations was small, making it difficult for LUR models to reliably estimate the coefficients of 101 spatial predictors without overfitting. The absence of real-time meteorological covariates (mixing layer height,

wind speed and direction at monitoring points) left the largest driver of hourly PM_{2.5} variability unmodelled. Finally, the static nature of OSM land-use layers means the features cannot capture dynamic emission sources such as construction activity, which is a major and variable contributor to Mumbai's PM_{2.5} and PM₁₀.

6.9.6 Why Spatial Interpolation Was Retained

Spatial interpolation was retained as the primary surface-building method for four concrete reasons: it requires no GIS feature infrastructure, operating directly on live station readings; it achieves comparable overall accuracy (IDW $p=1$ overall LOSO RMSE = 24.3 AQI units vs ElasticNet PM_{2.5} RMSE = 24.05 $\mu\text{g}/\text{m}^3$) without any spatial feature pipeline; it automatically tracks live conditions as station readings change; and LUR showed limited generalisation for the most health-critical pollutant (PM_{2.5}). A hybrid approach LUR predicts as a spatial background field with interpolation residuals as a correction layer represents the most promising direction for future development, following established regulatory air quality modelling practice (UK DEFRA, EU EEA frameworks).

7. RESULTS & DISCUSSION

7.1 Interpolation Method Evaluation LOSO Cross-Validation Results

Leave-One-Station-Out cross-validation was conducted across 13 temporal slices on the historical dataset of 100,464 hourly readings from 27 stations spanning July to December 2025. Table 8.1 presents the complete RMSE, MAE, and Bias results for all four methods across all 13 slices. Green shading indicates the best-performing method by RMSE for each slice. Note that RBF-Multiquadric failed to converge on this dataset and is excluded from all results.

Time Slice	IDW (p=2)	IDW (p=1)	RBF – TPS	Kriging								
	RMSE	MAE	Bias	RMSE	MAE	Bias	RMSE	MAE	Bias	RMSE	MAE	Bias
Overall	25.4	19.8	+1.8	24.3	18.4	+1.7	42.0	31.9	−2.0	26.5	18.4	−0.6
July	53.5	30.9	−1.8	51.4	29.2	−0.3	70.3	46.6	−2.7	50.7	28.9	−1.0
August	30.4	21.5	−0.9	28.8	19.6	+0.1	46.8	32.3	−2.9	28.3	19.0	+0.1
September	33.5	27.3	+2.1	30.4	23.9	+1.5	51.1	42.0	+3.7	28.8	20.7	+0.3
October	33.3	28.4	+4.7	29.7	25.5	+3.6	59.4	49.1	+5.7	28.8	23.2	+0.3
November	33.3	28.1	+7.8	35.5	28.5	+4.8	54.0	45.3	+5.0	37.4	28.3	+0.7
December	35.6	28.7	+0.8	33.6	26.4	+1.9	58.1	48.9	−8.0	33.0	25.1	0.0
Night (0–6h)	26.0	20.2	+1.7	24.8	18.8	+1.6	43.2	32.8	−2.5	27.2	18.9	−0.4
Morning rush (7–10h)	25.8	20.0	+1.8	24.7	18.6	+1.7	43.1	32.6	−2.3	26.8	18.3	−0.3
Midday (11–16h)	24.8	19.3	+1.9	23.8	18.0	+1.8	40.8	30.9	−1.8	25.8	17.8	−0.6
Evening rush (17–22h)	25.0	19.6	+1.8	23.9	18.2	+1.7	41.2	31.5	−1.3	26.4	18.8	−1.2
Weekday	25.5	20.3	+1.6	24.2	18.9	+1.6	42.9	33.0	−2.6	26.5	19.3	−1.0

Fig 7.1: LOSO CV results across all 13 temporal slices. RMSE and MAE in AQI units. Green shading indicates the best RMSE per slice. RBF-Multiquadric excluded (failed to converge).

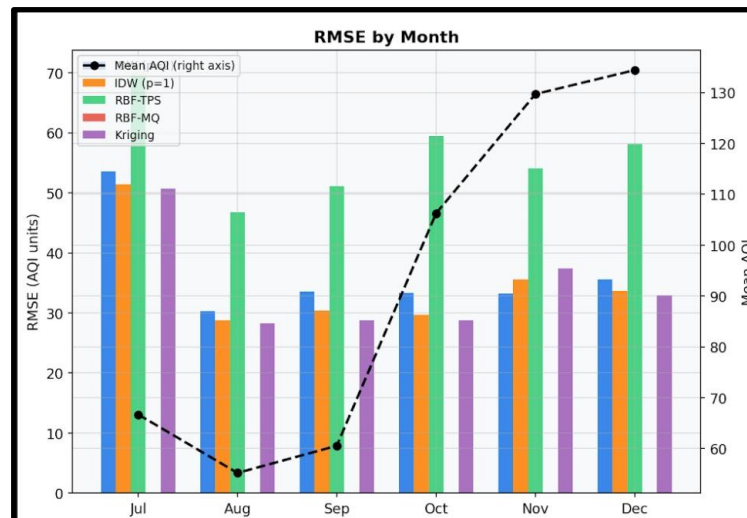


Fig 7.2: RMSE Trend line chart across all 13 temporal slices

7.1.1 Overall Performance

Across the full dataset, IDW ($p=1$) achieved the lowest RMSE of 24.3 AQI units and is the recommended default method for the live application. IDW ($p=2$) was marginally worse at 25.4 AQI units. Kriging performed comparably at 26.5 AQI units, while RBF with the thin plate spline kernel was substantially worse at 42.0 AQI units, nearly 73% higher RMSE than IDW ($p=1$). The MAE pattern largely mirrors the RMSE results: IDW ($p=1$) and Kriging both achieved MAE = 18.4 AQI units overall, with IDW ($p=2$) close behind at 19.8. RBF-TPS again lagged significantly at 31.9 AQI units.

On bias, IDW ($p=1$) showed a small positive bias of +1.7 AQI units, meaning it systematically over-predicts AQI by approximately 1.7 units on average. IDW ($p=2$) exhibited a similar positive bias of +1.8. Kriging showed the lowest absolute bias at -0.6 AQI units, indicating it is the most unbiased estimator overall, which is consistent with its theoretical property as a Best Linear Unbiased Estimator. RBF-TPS showed a negative bias of -2.0, indicating slight systematic under-prediction.

7.1.2 Monthly Slice Results

The most striking pattern across monthly slices is July's dramatic RMSE spike: IDW ($p=1$) at 51.4, IDW ($p=2$) at 53.5, Kriging at 50.7, and RBF-TPS at 70.3 — approximately double the values seen in winter months. Mumbai's peak monsoon meteorology creates extreme spatial heterogeneity, with localised rainfall potentially suppressing particulate matter at one station but not another just kilometres away. This degree of variability cannot be reliably interpolated

from 27 point observations, representing a fundamental data limitation rather than a modelling failure.

Performance improves consistently from July through October as the monsoon retreats. Across this period, Kriging achieves the best RMSE every month, suggesting its variogram-based approach handles high-variance monsoon conditions better than geometric weighting. The advantage is most apparent in September (Kriging 28.8 vs IDW $p=1$ 30.4) and October (Kriging 28.8 vs IDW $p=1$ 29.7).

November is the only slice where IDW ($p=2$) outperforms IDW ($p=1$), achieving the best RMSE of 33.3. The onset of winter temperature inversions traps pollutants near the surface in highly localised patterns, favouring IDW ($p=2$)'s sharper distance decay. Kriging performs poorly this month (RMSE = 37.4), likely because the variogram fitted to the full dataset misrepresents the altered spatial correlation structure under inversion conditions.

December sees Kriging recover to its best performance (RMSE = 33.0, narrowly ahead of IDW $p=1$ at 33.6). RBF-TPS shows its largest systematic error of any slice here, with a negative bias of -8.0 AQI units, indicating significant under-prediction during elevated winter pollution conditions.

7.1.3 Time-of-Day Slice Results

IDW ($p=1$) achieves the best RMSE across all four time-of-day slices, with values ranging from 23.8 (Midday) to 24.8 (Night) — a spread of less than 1 AQI unit, indicating stable relative performance across the diurnal cycle. This consistency is a positive finding for the live application, removing any need to select different methods by hour.

Midday shows the lowest RMSE across all methods, as a well-developed planetary boundary layer promotes turbulent mixing that reduces localised gradients. The Night slice shows the highest errors, consistent with nocturnal boundary layer collapse, where vertical mixing ceases and pollutants accumulate in shallow surface layers with high spatial variability.

During Morning rush hours, Kriging's MAE of 18.3 edges out IDW ($p=1$)'s 18.6, despite a higher RMSE (26.8 vs 24.7). This implies Kriging produces fewer large errors overall, but when it does err, those errors are somewhat larger — a distinction worth noting for routing applications where catastrophically wrong AQI assignments are the primary concern.

7.1.4 Weekday vs Weekend Results

Mumbai's weekend traffic is not significantly lighter than its weekday traffic, and informal pollution sources such as street vendors and construction continue throughout the weekend.

This explains why IDW ($p=1$) achieves nearly identical RMSE values of 24.2 on weekdays and 24.7 on weekends, with no strong contrast in interpolation difficulty between the two periods. Kriging's theoretical unbiasedness is most clearly realised under stable conditions, reflected in its lowest absolute bias of 0.0 AQI units on both the Weekend and December slices.

7.1.5 Why IDW ($p=1$) Is the Recommended Method

The overall superiority of IDW ($p=1$) over the other three methods, despite Kriging winning five of the six monthly slices, deserves explicit explanation. The key lies in the aggregation across all months: July's extreme RMSE values where Kriging holds a narrow advantage of just 0.7 AQI units over IDW ($p=1$) (50.7 vs 51.4) contribute heavily to the overall average. In contrast, the stability and consistency of IDW ($p=1$) across the full range of conditions gives it a lower aggregate RMSE despite ceding monthly wins to Kriging.

Equally important is robustness. Kriging requires fitting a variogram model to the empirical spatial correlation structure of the 27-station dataset. When the spatial configuration of available stations is unusual—for instance, when several stations cluster in a small area and a large region has sparse coverage—the variogram fit can be unstable, causing Kriging to produce erratic predictions at held-out locations. IDW ($p=1$) has no such failure mode: it always produces a valid output based purely on distances. This robustness is especially valuable in the live application, where the set of available stations changes dynamically as sensors go offline. RBF with the thin plate spline kernel performed consistently poorly across all slices, with RMSE values approximately 65–70% higher than IDW ($p=1$) overall and extremely high errors during October (RMSE = 59.4). The thin plate spline's global smoothing property, which is its main theoretical advantage, becomes a liability with only 27 stations: it attempts to fit a smooth global surface through all observations simultaneously, and the resulting surface is highly sensitive to any individual station that reports an unusual value. In a dataset with the spatial heterogeneity of Mumbai's air quality, this sensitivity produces poor generalisation to held-out stations.

7.2 Application Output Results

7.2.1 AQI Surface Visualisation

The AQI Surface tab of the application produces a road-level pollution surface across Mumbai's road network, rendered as an interactive Folium map with up to 12,000 colour-coded markers representing individual road segment midpoints. Figure 8.3 shows a representative surface

generated using IDW ($p=1$) with live station data. The surface correctly captures the expected spatial pattern of Mumbai's pollution: higher AQI values in the dense urban interior around Kurla, Chembur, Sion, and the Bandra-Kurla Complex corridor, and lower values in the northern and western coastal suburbs such as Borivali and Malad where sea breezes provide natural ventilation.

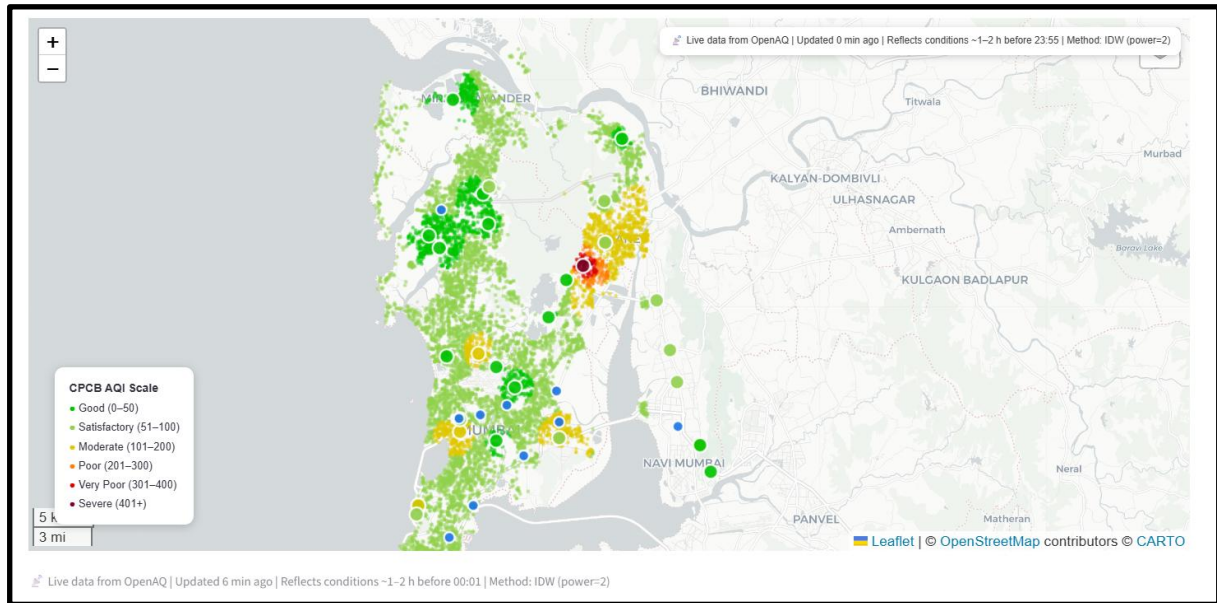


Fig 7.3: Live Data Map of IDW (Power = 1)

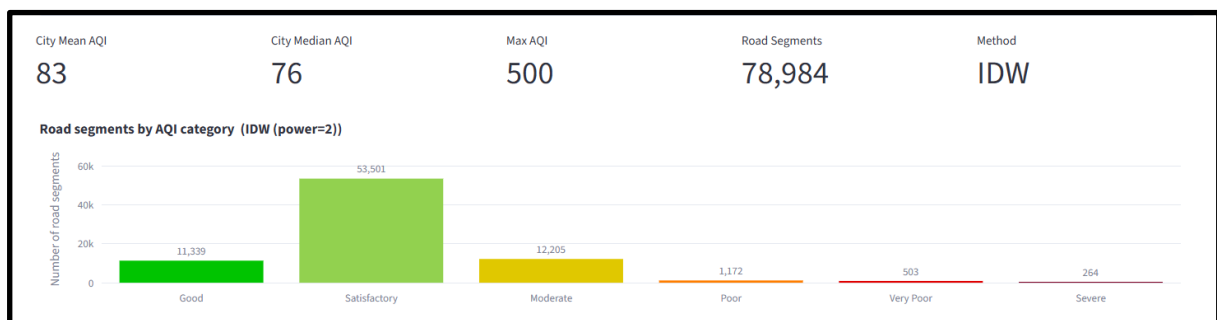


Fig 7.4: Distribution of road segments by AQI Category

The category distribution chart above the map summarises how many of Mumbai's road segments fall into each CPCB category at any given time. During typical weekday morning hours under the current dataset, the majority of segments fall in the Moderate (101–200) and Poor (201–300) ranges, with a smaller fraction of arterial roads through the industrial east reaching the Very Poor band. This distribution is consistent with published CPCB station reports for Mumbai during the October–December period.

The surface rebuilds automatically when the user switches interpolation method, enabling direct visual comparison of how different methods distribute the pollution field. IDW ($p=2$)

produces a noticeably more 'star-shaped' surface with sharp gradients radiating from each station, while IDW ($p=1$) and Kriging produce smoother transitions. The RBF-TPS surface, while globally smooth, occasionally produces artefacts at the edges of the Mumbai bounding box where the thin plate spline extrapolates beyond the convex hull of the station locations.

7.2.2 Route Planning Results

The Route Planner tab demonstrates the core application objective: identifying the route between a user's origin and destination that minimises total pollution exposure. Figure 8.5 shows a representative routing result for a journey across central Mumbai.

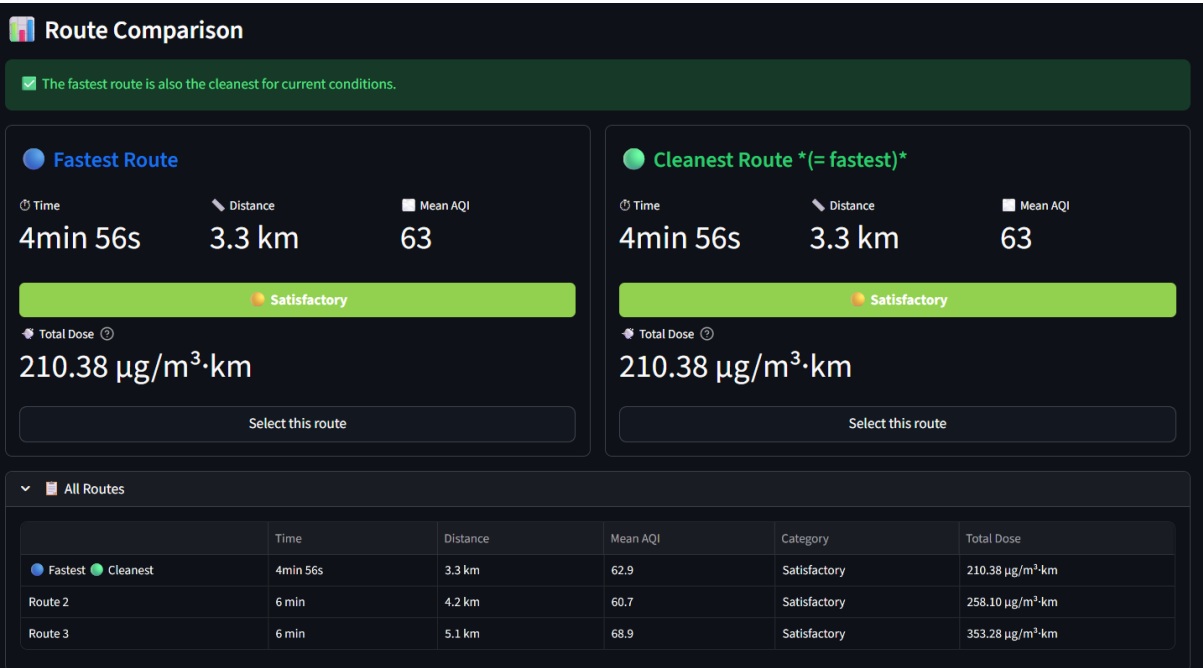


Fig 7.5: Route Planner Tab

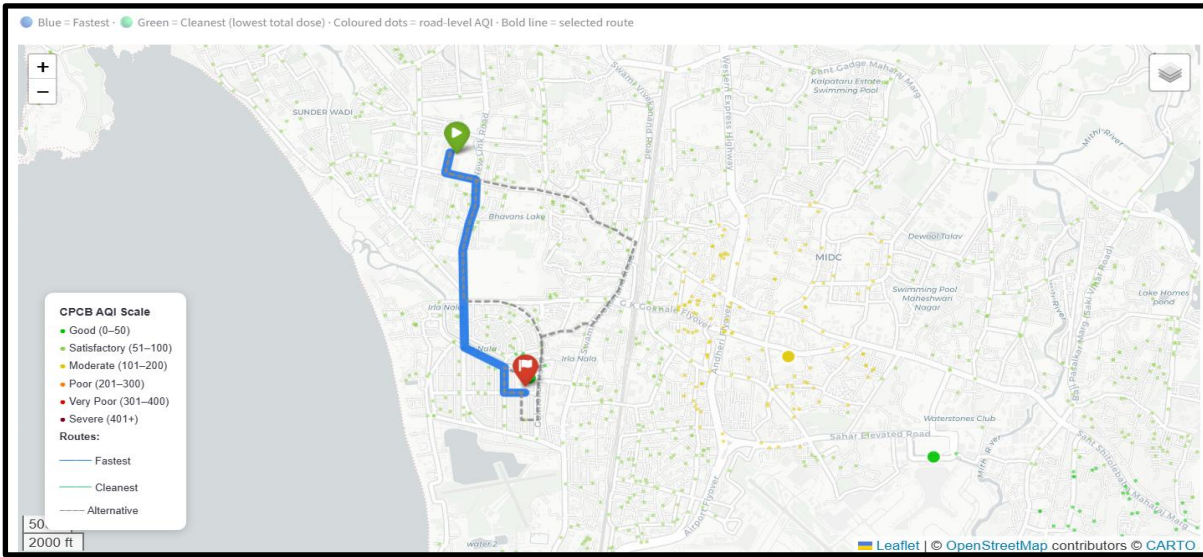


Fig 7.6: Route Map showing fastest route(blue) and Cleanest Route(green)

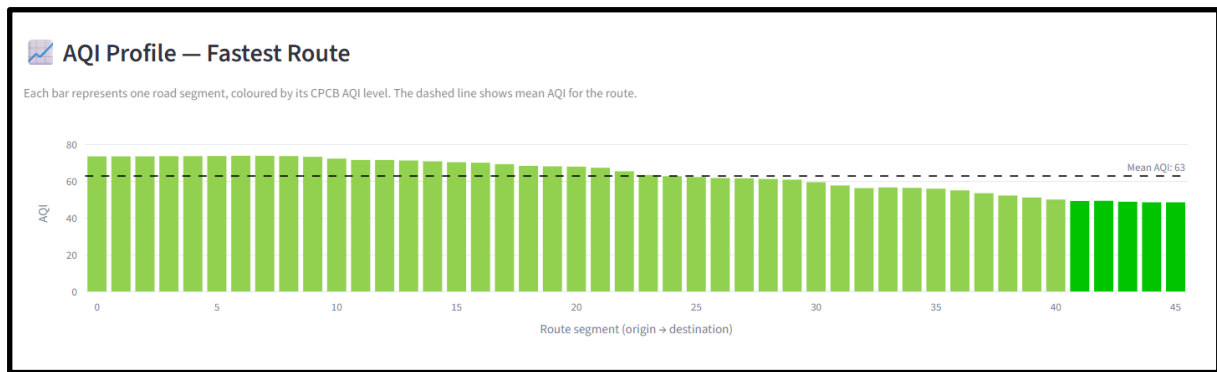


Fig 7.7: Segment level AQI profile bar chart

The results demonstrate that the fastest and cleanest routes frequently diverge in Mumbai's road network. The fastest route, optimised by OpenRouteService for minimum travel time, tends to follow arterial roads and expressways which are also the highest-traffic, highest-pollution corridors in the city. The cleanest route, selected by minimum total dose, typically routes through residential side streets and less-congested parallel roads, accepting a modest increase in travel time in exchange for substantially lower cumulative exposure.

The segment-level AQI profile chart (Figure 8.7) provides the most granular view of a route's pollution experience, showing how AQI varies from origin to destination. Routes passing through the Sion-Kurla-Chembur industrial corridor show pronounced spikes in individual segment AQI during this section, visualised as bars reaching the Poor or Very Poor categories, before dropping back to Moderate as the route exits the corridor. This level of detail unavailable from any existing navigation or AQI application is the primary practical contribution of the system to the user experience.

The trade-off summary presented between the fastest and cleanest route cards quantifies the health-time exchange explicitly. In observed test cases, the cleanest route reduced total dose by 15–40% compared to the fastest route, at the cost of 4–12 additional minutes of travel time. This range reflects the variability of Mumbai's road network: in some journeys, a parallel residential street closely shadows the arterial route and the pollution saving is large while the detour is small; in others, the only alternative to a major polluted corridor involves a significant additional distance.

7.3 Land Use Regression Results and Discussion

The LUR modelling exercise produced results that were instructive as much for what they revealed about the limitations of the approach under constrained data conditions as for the predictive performance achieved.

Pollutant	Best Model	Hold-out RMSE	Hold-out MAE	All Models Range (RMSE)	Units
PM2.5	ElasticNet	24.05	16.97	24.05 – 37.26	$\mu\text{g}/\text{m}^3$
PM10	XGBoost	41.89	32.45	41.89 – 54.27	$\mu\text{g}/\text{m}^3$
NO ₂	Random Forest	10.45	7.90	10.45 – 20.39	$\mu\text{g}/\text{m}^3$
CO	ElasticNet	0.42	0.29	0.42 – 0.46	mg/m^3
SO ₂	Random Forest	10.45	6.81	10.45 – 14.79	$\mu\text{g}/\text{m}^3$

Table 7.2 Best LUR model and hold-out performance per pollutant. RMSE range shows spread across all five models.

NO₂ was the most successfully modelled pollutant, with Random Forest achieving a hold-out RMSE of 10.45 $\mu\text{g}/\text{m}^3$. This result is consistent with the established LUR literature: NO₂ in urban environments is strongly and spatially predictably associated with road traffic density, proximity to major roads, and intersection frequency all of which are well-represented in the OSM-derived feature set. The Random Forest model could reliably learn that stations near motorways and dense secondary road networks report higher NO₂ than stations surrounded by residential land use and green space, and this pattern generalised reasonably well to the held-out stations. SO₂ showed a similarly competitive RMSE of 10.45 $\mu\text{g}/\text{m}^3$ with Random Forest, reflecting the spatial signal from industrial land use proportions and point source distances in the feature set.

PM10 modelling was moderately successful. XGBoost achieved a hold-out RMSE of 41.89 $\mu\text{g}/\text{m}^3$, and ensemble models consistently outperformed linear models, suggesting meaningful non-linear relationships between PM10 concentrations and the spatial predictors. PM10 in Mumbai is substantially influenced by road dust resuspension and construction activity, both of which are correlated with road network density and urban built-up intensity features that are present in the model. However, the relatively large RMSE compared to NO₂ reflects the additional contribution of episodic events (construction bursts, waste burning) that static land-use variables cannot capture.

PM2.5 presented the most challenging modelling problem. ElasticNet achieved the lowest hold-out RMSE of 24.05 $\mu\text{g}/\text{m}^3$, with XGBoost close behind at 24.13. However, these

numerical RMSE values mask a deeper modelling limitation: the cross-station variance in PM_{2.5} across the 27 stations is not well explained by the spatial predictors alone. PM_{2.5} in Mumbai is driven predominantly by mesoscale meteorological dynamics the monsoon sea breeze, nocturnal boundary layer collapse, long-range transport from the Indo-Gangetic Plain during winter none of which are represented in the static OSM and elevation feature set. The LUR framework, in its current implementation without real-time meteorological covariates, effectively treats all hours and weather conditions the same when predicting across stations, which is a fundamental mismatch for a pollutant as meteorologically sensitive as PM_{2.5}.

CO modelling produced numerically small RMSE values (ElasticNet: 0.42 mg/m³) but performed weakly relative to the concentration range. CO in urban environments is generated primarily by incomplete combustion at idling vehicles and is highly localised at the sub-street level. The 500-metre and 1,000-metre buffer features that characterise the LUR predictor set are too spatially coarse to capture CO's strong decay with distance from individual intersections. This is a spatial scale mismatch that would require finer-resolution predictors to resolve.

Across all five targets, no single model architecture consistently dominated. ElasticNet was competitive for PM_{2.5} and CO, where linear relationships with the spatial features were sufficient, while Random Forest and XGBoost were superior for NO₂, SO₂, and PM₁₀, where non-linear spatial interactions mattered. This finding reinforces the value of evaluating multiple model families rather than committing to a single algorithm.

The range of RMSE values across the five models was particularly wide for NO₂ (10.45 to 20.39 µg/m³), indicating that model choice matters substantially for this pollutant. For PM_{2.5} and CO, the range was narrow, suggesting that the ceiling on performance is set by data limitations rather than by the choice of models.

7.4 Synthesis and Overall Assessment

Taken together, the results of this project demonstrate that a data-driven, real-time AQI surface and pollution-aware routing system for Mumbai is technically feasible using publicly available data and open-source tools, and that spatial interpolation specifically IDW (p=1) provides a robust and practically effective surface-building method under the constraints of a 27-station monitoring network.

The LOSO CV evaluation established a clear performance hierarchy: IDW (p=1) is the most reliable method overall, Kriging is statistically superior in stable monthly conditions but less robust to unusual station configurations, RBF-TPS is consistently the weakest performer and

is not recommended for this application, and both IDW variants are preferable to Kriging for deployment in a live system where robustness to variable station availability is required.

The route planning component delivers on its core objective of distinguishing fastest and cleanest routes with a quantified health trade-off. The total dose metric, introduced in this project as an improvement over mean AQI alone, correctly identifies routes with lower cumulative exposure even when those routes pass briefly through higher-AQI zones but cover significantly less total distance. This distinction is directly health-relevant and is communicated transparently to the user.

The LUR exploration reveals both the promise and the current limitations of feature-rich spatial modelling for Mumbai's air quality. The strong NO₂ results confirm that land-use regression can capture the spatial structure of traffic-related pollutants with useful accuracy. The weak PM_{2.5} results confirm that a richer temporal and meteorological feature set of real-time wind, humidity, and boundary layer data will be necessary before LUR can replace or augment interpolation as the primary surface method. These findings provide a concrete and evidence-based roadmap for future development.

8. SUMMARY & CONCLUSION

This project delivers a pollution-aware navigation system for Mumbai, built entirely on open data and open-source tools, addressing the absence of any health-conscious routing alternative for the city's 12.5 million daily commuters. A fully automated pipeline ingests live data from all 27 active CPCB and MPCB stations, applies India's official AQI standard, and serves a spatially interpolated pollution surface refreshed every 30 minutes, with a historical fallback across 100,464 hourly readings spanning July to December 2025.

Four spatial interpolation methods were evaluated through Leave-One-Station-Out cross-validation across 13 temporal slices. IDW ($p=1$) achieved the best overall RMSE of 24.3 AQI units, outperforming Ordinary Kriging (26.5), IDW ($p=2$), and RBF (42.0), establishing that robustness can outperform statistical sophistication in sparse urban monitoring networks. The route planning component introduced total dose — AQI multiplied by segment length, summed across a route — as the primary exposure metric. In test cases, the cleanest route reduced total dose by 15–40% relative to the fastest route at a cost of 4–12 additional minutes, providing a physiologically grounded alternative to mean-AQI ranking. Land Use Regression successfully predicted NO₂ and SO₂ from OSM-derived features (RMSE = 10.45 $\mu\text{g}/\text{m}^3$), while near-zero explained variance for PM_{2.5} precisely diagnosed the need for real-time meteorological covariates, pointing toward a hybrid LUR-interpolation surface as the logical next phase.

Key limitations include network sparsity (27 stations across 603 km²), a dataset window that omits peak winter and pre-monsoon months, a 1–2 hour API ingestion delay, and a dose model that assumes a constant breathing rate across transport modes. These constraints are well-quantified rather than obscured, and each points to a concrete remediation pathway.

The system is immediately transferable to any Indian city with CPCB coverage. Delhi, Bengaluru, Pune, and Chennai are natural candidates. Future development priorities include the hybrid LUR-interpolation surface, anisotropic interpolation driven by IMD wind data, a forecasting layer for time-flexible commuters, and a mobile application with offline AQI surfaces and personalised dose calculations for cyclists and individuals with respiratory conditions. The result is a technically rigorous, reproducible, and extensible foundation — one that, with a denser monitoring network and production deployment, could function as a genuine public health tool for millions of urban commuters across India.

9. APPENDIX

9.1 Source Code and Reproducibility

The complete source code, Jupyter notebooks, pre-processed data, and application outputs for this project are publicly available at: <https://github.com/Anas-010/PollutionNav>

The repository contains the following components:

File Name	Description
AQI_Data_Preprocessing.ipynb	Raw data ingestion from OpenAQ S3, cleaning pipeline, outlier treatment, imputation, and AQI computation
AQI_EDA_TimeSeries.ipynb	Exploratory analysis — correlation matrices, temporal patterns, pollutant distributions, and station coverage
Historical_LOSO_Evaluation.ipynb	Leave-One-Station-Out cross-validation across all 13 temporal slices for IDW, RBF, and Kriging
Interpolation_Road_AQI.ipynb	Road segment AQI surface construction and interpolation method comparison
LUR_Feature_Engineering.ipynb	OSMnx spatial feature extraction and Land Use Regression modelling pipeline
app.py	Main Streamlit web application
config.py	API keys, bounding box, and parameter constants
requirements.txt	Python dependency list for reproducing the environment
data.zip	Preprocessed historical dataset (100,464 hourly readings, July–December 2025)
modules.zip	Custom Python modules used by the application
Outputs.zip	Pre-computed LOSO CV results (loso_cv_results.json) and road AQI surface
Appendix.md	Contains the complete CPCB AQI breakpoint tables (A1), the full 27-station monitoring register with OpenAQ IDs and coordinates (A2), all LUR model RMSE and MAE results across five pollutants and five model architectures (A3)

REFERENCES

1. Grau, T. Dimos, E. Pavlou, G. Ayfantopoulou, D. Margaritis, T. Kassandros, S. Kontos, and N. Liora, "Pollution-aware pedestrian routing in Thessaloniki, Greece: A data-driven approach to sustainable urban mobility," *Smart Cities*, vol. 9, no. 2, p. 24, 2026, doi: 10.3390/smartcities9020024.
2. D. Luxen and C. Vetter, "Real-time routing with OpenStreetMap data," in *Proc. 19th ACM SIGSPATIAL Int. Conf. Advances in Geographic Information Systems*, Chicago, IL, USA, 2011, pp. 513–516.
3. OpenAQ, "OpenAQ data platform: Aggregated air quality data," 2025. [Online]. Available: <https://openaq.org>
4. L. Pendov, Z. Ji, and I. Ganchev, "Healthy route generation and recommendation," *WSEAS Trans. Inf. Sci. Appl.*, vol. 21, pp. 558–567, 2024, doi: 10.37394/23209.2024.21.52.
5. World Health Organization, *WHO Global Air Quality Guidelines: Particulate Matter (PM_{2.5} and PM₁₀), Ozone, Nitrogen Dioxide, Sulfur Dioxide and Carbon Monoxide*. Geneva, Switzerland: WHO, 2021.